Day 1 Presentation

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Introduction to the Course

Our class has several goals:

- Mathematical prerequisites for first year of methods courses.
- Introduction to probability and statistical inference.
- Introduction to computational methods.

Goals of statistical inference/probability

- Generate knowledge on the basis of observed data.
- The goal is to make claims about unobserved phenomena, often referred to as a theoretical **population**.
- The phenomena may be unobserved because
  - it is in the future (forecasting)
  - outside the sample (extrapolation)
  - not directly measured (latent)
- A special case is when we want to predict counterfactual outcomes (also called causal inference).

Advantages of statistics

- Make explicit our assumptions.
- We learn non-obvious properties of inferences.
- We get a measure of the degree of certainty in our inferences.

Toolbox: Probability theory

- The second half of the course begins with probability, goes through statistical inference.
- This is a necessary prerequisit for linear models, the workhorse of quantitative work.

Connections between subjects

- Set theory + Logic : Analytic political theory, programming, almost everything.
- Calculus : a p-value is the solution to an integral, the M in MLE, game theory.
- Linear algebra : basis of linear regression.
- Mathematical thinking : Everything.

R + RStudio

- For this class, we will use R, but interact with it using RStudio.
  - R is a flexible language for data analysis.
  - RStudio is an IDE, or ***Integrated Development Environment***, which offers an interface to R.
- Your assignments can all be turned in pdf format using Latex + Rmarkdown, which is built into RStudio.

Principles to keep in mind

- You can learn this, it will take time and practice.
- You will not break anything important.
- It is fun.

Rstudio Panes

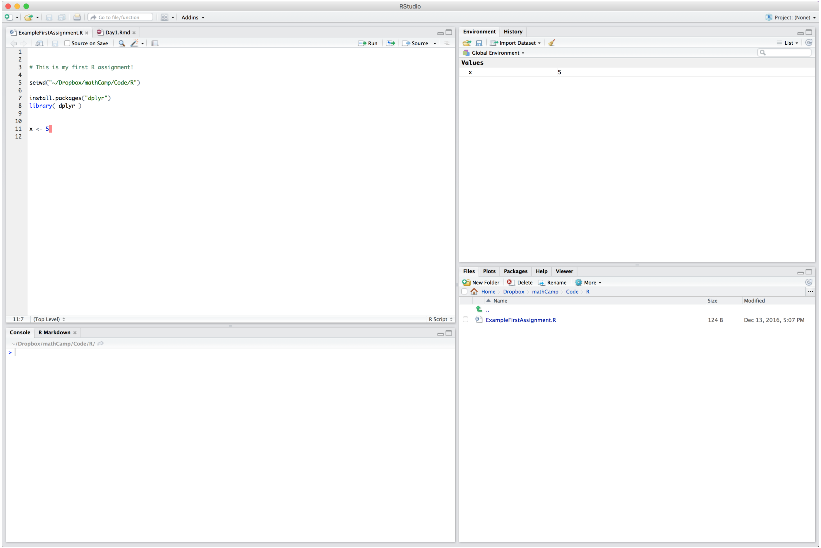

Figure 1: RStudio In Action

Source: Where your code is edited.

The top left part of the screen is the Source.

- This is where you type in your code, in what is called a script.
- The default document is a .R document (plain text).
- The more advanced documents are called Rmarkdown documents, .Rmd
- RStudio highlights syntax and autocompletes code (tab)

Console: Where your code is run.

The bottom left part of the screen is the Console.

- This is an interactive code environment.
- The little $>$ is where the code runs.
- A + means that it is waiting for a command. `esc` to cancel.
- control - l to clear the Console.

Environment/History

The top right part of the screen has the Environment and History tabs.

- Environment includes all the objects in RAM.
- As you load or create data, this populates your Environment
- The little grid icon to the right of Data objects opens a "view" screen.
- `ls()` returns all the objects in the global environment.
- `rm(x)` removes an object x.

Objects and Functions in R

- There are several fundamental datatypes, which you will learn.
  - Characters: strings enclosed in `"Colorado"`
  - Numeric: integers and doubles `21.1`, `4L`,`9e2`
  - Logical: `TRUE`, `FALSE`, `NA`
- You can learn about the type of an object by using `str()`.
- Functions have a name, followed by ()

```
rnorm(10)
```

```
##  [1]  1.74707308  2.05539477 -0.40947395
##  [4]  0.25823075  1.13582702  0.86286772
##  [7] -0.10397564  0.43826618  1.08578834
## [10] -0.08132534
```

Useful Shortcuts

Purpose	Keyboard shortcut	Menu Path
Run line	command - enter	Code / Run Selected Line(s)
Code Completion	tab	
Clear Console	control - l	Edit / clear console
Save Script	command - s	File / save
Reformat Script	command - a	Code / Reformat Code
list shortcuts	option - shift - k	Help / Keyboard Shortcuts

Workflow

Set-up for assignments in Rmarkdown

- R markdown documents are created in the File menu and appears in the Console.
- We have provided templates, with lots of code examples.
- R markdown has embedded **r code** and embedded **latex** code.

LaTeX Code

- LaTeX was invented to typeset mathematics:

```
$\int_{i=1}^\infty \frac{1}{x} = \delta$
```

gives you

$\int_{i=1}^\infty \frac{1}{x} = \delta$

- There is a template sheet available on Canvas with many useful commands.

How to get Help

Help in R

- `?` + the command. eg. `?plot`, `?'=='`,`?'%in%'` loads help for **loaded** functions.
- help(package="dplyr") gives the main page for a package.
- `??` + command searches R help pages.

How to google it.

- Add CRAN + google search term
- Add [r] + google search term
- Go to datacamp.com, stackoverflow

How ask for help online.

- Isolate one issue. Each issue gets it own minimal complete verifiable example.
- Try to make the mistake happen in as few lines of code as possible (minimal/verifiable).
- Then send the whole script (complete).

R session.

- This Friday morning at 9:30 we will have a session in R going through things in detail.
- Rather than your normal Lab space, Wieboldt Hall 130, we will meet in SHFE 146 (the econ building).